

Tarandeep Singh Khurana

(+91) 8989244098

Email

LinkedIn

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Portfolio

SKILLS

Programming Skills: Python, Java, SQL

Databases: PostgreSQL (Azure – Flexible Server), MySQL, Supabase, Vector Databases (Pinecone, FAISS, ChromaDB)

Libraries & Frameworks: LangChain, LangGraph, FastAPI, Flask, Hugging Face, Pandas, NumPy, Scikit-Learn, Keras, PyTorch, Matplotlib, Seaborn

Cloud & Infrastructure: Azure (VMs, Blob Storage, PostgreSQL – Flexible Server, Azure Foundry), Docker, GitHub Actions, Hostinger VPS

Developer Tools/Platforms: VS Code, GitHub, Postman, LangSmith, Langfuse, Jupyter Notebook, Google Colab, MLflow

Core Concepts: LLM Workflows, Agentic AI, Generative AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Engineering, NLP, Deep Learning, Machine Learning, Data Science, MLOps, EDA, OOP, DSA, Scrum

Domain Knowledge: Quantitative Finance, Algorithmic Trading, Financial Markets (Equities, F&O, Commodities — NSE/BSE/MCX), Technical Analysis (OHLCV, Candlestick Patterns, Indicators), Market Data Systems (Live Quotes, Gainers/Losers, Market Breadth), Backtesting Workflows, LLM-powered Financial Analytics, Semantic Search on Financial Data (News, Announcements), Event-Driven Data Pipelines

WORK EXPERIENCE

Software Engineer Intern, Astryx, Bengaluru, Karnataka

October 2025 – April 2026

- Architected a **multi-agent LangGraph system** using an **LLM-based semantic router** with **4 nodes (Router, Handler, Executor, Synthesizer)** and **8 specialized skills** — market data, NL-to-SQL-driven stock screening and quantitative analysis, announcement search, news search, web research, chart visualization, scheduling and backtesting algorithmic strategy — **routing queries across simple and complex skill-level execution paths, with 40+ tools orchestrated across specialized skills.**
- Designed and implemented **short-term memory via progressive conversation summarization**, preserving context across multi-turn interactions while **optimizing token efficiency** and **maintaining coherent agent reasoning.**
- Designed and refined **structured prompts** across **router, skill agents, and synthesizer**, incorporating **few-shot prompting, chain-of-thought guidance, and operational guardrails** to enforce **role-specific behavior, control tool usage, improve routing accuracy, reduce hallucinations, and ensure consistent multi-step execution.**
- Built **6 end-to-end financial data ingestion pipelines** for NSE/BSE — **daily OHLCV data, company master data, corporate filings, and a concurrent async news scraper fleet of 58 scrapers across 15+ Indian financial publishers** (Economic Times, Moneycontrol, Mint, Reuters India, CNBC TV18, and more) — with URL-hash deduplication, OG metadata enrichment, embedding generation, batch upserts, and GitHub Actions orchestration (filings every 30 min, news every 20 min, company updates and OHLCV on weekdays) using **Azure-hosted data services.**
- Designed a **two-tier semantic search system over 17M+ announcement chunks (BSE 11.7M, NSE 5.9M)** using **text-embedding-3-large (3072-dim)**, PostgreSQL halfvec with **HNSW indexes**, and **Azure Document Intelligence** for **PDF OCR and chunk extraction.**
- Engineered a **multi-provider real-time market data layer** aggregating Upstox, Dhan, Yahoo Finance, and NSE/BSE scrapers — with priority-based merging, retry logic, and fallback chains — covering equities, F&O, commodities, currencies, and indices across NSE/BSE/MCX.
- Implemented **secure LLM-database integration** with **read-only SQL enforcement, auto-LIMIT injection, parallel query execution** via ThreadPoolExecutor, and **natural language to PostgreSQL generation** — **enabling quantitative stock screening and technical pattern detection** over historical OHLCV data (breakouts, volume spikes, momentum, sector rankings, index constituents, multi-criteria).
- Built a **streaming response architecture** using **OpenAI Responses API** — **emitting thinking tokens, answer tokens, progress events, and structured content (stock cards, charts, schedule widgets)** via thread-local stream context management — **delivering real-time visibility into agent reasoning, tool calls, and intermediate steps to keep users engaged throughout long-running financial queries.**
- Developed a **model factory with instance caching** by (model, temperature, streaming) signature and startup warmup to eliminate cold-start latency — **architected deliberate model tiering across all skills**, using **GPT-5-mini** for routing, skill execution, and synthesis on simple paths, while reserving **GPT-5.4** only for NL-to-SQL generation, backtesting, and complex-path synthesis — **cutting inference costs significantly while maintaining response quality where it mattered.**
- Built and integrated **backend APIs in Node.js** alongside Python microservices, **collaborating closely with frontend teams on UX flows**, system contracts, and real-time data presentation — and **authored a detailed design spec for the backtesting form**, defining component layout, parameter structure, and interaction flow for strategy configuration interface.

AI/ML Intern, H TO H LLP, Noida, Uttar Pradesh

July 2025 – September 2025

- Built and deployed a **Retrieval-Augmented Generation (RAG) system** on **Hostinger VPS**, running 24/7 in real-time to fetch and analyze stock market news. Integrated **LangChain, OpenAI LLMs, FAISS, yfinance, and web scraping** to deliver contextual insights (signal, potential, confidence, sector analysis) via **Telegram.**
- Developed and deployed **algorithmic trading workflows** on **VPS** using **Python**. Integrated **ICICI Breeze API, Zerodha KiteConnect API, and TradingView** for **live execution and strategy automation.**
- Designed and **automated a technical analysis system (support/resistance, consolidation, breakout detection)** using **intraday, daily, and weekly OHLCV data**, with results logged in **Google Sheets** for continuous monitoring.
- Evaluated **48 candidate submissions** for algorithmic trading systems, ensuring quality, correctness, and adherence to specifications.

PROJECTS

Stock News Analysis & Market Signal Generator | Python, LangChain, OpenAI API, FAISS, Hostinger VPS

[GitHub](#)

- Built a real-time **Retrieval-Augmented Generation (RAG)** system to analyze stock market news and provide actionable insights (signal, potential, confidence, sector analysis).
- Automated news ingestion via web scraping (**BeautifulSoup, Selenium**) and RSS feeds; orchestrated LLM processing with **LangChain** and **OpenAI GPT-4o-mini.**
- Stored and retrieved embeddings using **FAISS** with a **time-weighted retriever**; enriched insights with **Yahoo Finance** data and delivered via **Telegram bot.**

Joblens AI ChatBot | Python, FastAPI, LangChain, Gemini API, Streamlit, ChromaDB

[GitHub](#)

- Developed **JobLens**, an AI-powered conversational job search assistant using **RAG (LangChain + Google Gemini 2.5 Pro)** with contextual query handling.
- Automated real-time job data ingestion by scraping **Jobyaari (Selenium, BeautifulSoup)** and storing embeddings in **ChromaDB** updated every 20 minutes.

EDUCATION

SGSITS, Indore, Madhya Pradesh | B.Tech in Computer Science | CGPA: 8.99

2023-2027

Ratlam Public School, Ratlam, Madhya Pradesh | 12th CBSE | 93.8%

2022-2023

Shri Guru Tegh Bahadur Academy, Ratlam, Madhya Pradesh | 10th CBSE | 96.4%

2020-2021